

SINGLE-SUPPLY MICROPHONE AMPLIFIER CIRCUIT PROJECT DEMO

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Defective Single-Supply Microphone Amplifier

- A microphone amplifier using a single op-amp
- Original purpose: amplify small AC microphone signals (~ 10 mV)
- Circuit behaviour:
 - AC coupling
 - Intended to be a non-inverting op-amp
 - $A_v = 1 + \frac{100\text{ k}}{1\text{ k}} = 101 \left[\frac{\text{V}}{\text{V}} \right]$
 - Single-supply operations (negative rail = 0 V)

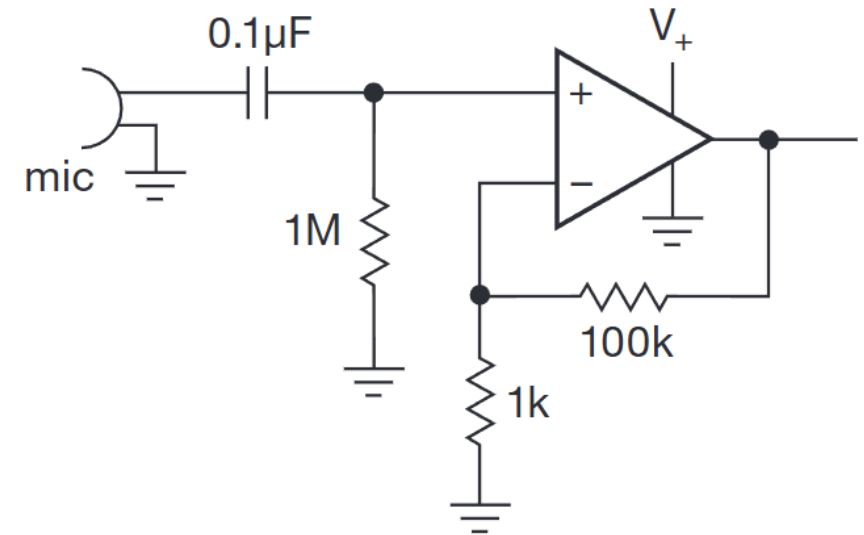


Fig. 1. Defective single-supply microphone amplifier

Defective Single-Supply Microphone Amplifier - Problem

- Capacitive coupling $\rightarrow$ DC at the $V_{in+} = 0\text{ V}$
- Voltage divider at 1 k & 100 k feedback resistors $\rightarrow V_{in-} \approx 0\text{ V}$
- Output (amplified signal) requires a negative output swing
 - Input mic signal = $\pm 5\text{ mV}$
 - Gain = 101
 - Output tries to swing around $\pm 0.5\text{ V}$
 - Negative power rail is restricted to 0 V
 $\rightarrow$ bottom half of wave is clipped
- LF411 data sheet: $+3.5\text{ V}$ to $+11.5\text{ V}$



Fig. 2. Output signal of the defective circuit

Solution – Biasing the Circuit to a Reference Voltage

- Objective: shift (bias) the AC input signal to mid-rail reference voltage so the amplified output stays within the supply rails
- Method: Voltage Divider of R1 & R2
 - $V_{in+} \text{ (or } V_{ref}) = V_+ \cdot \frac{R2}{R1+R2} = \frac{1}{2}V_+$
 - Note: $R1 = R2$
- With $V_+ = 12\text{ V}$, biasing shifts the AC signal to swing around 6V → valid range within the supply rails

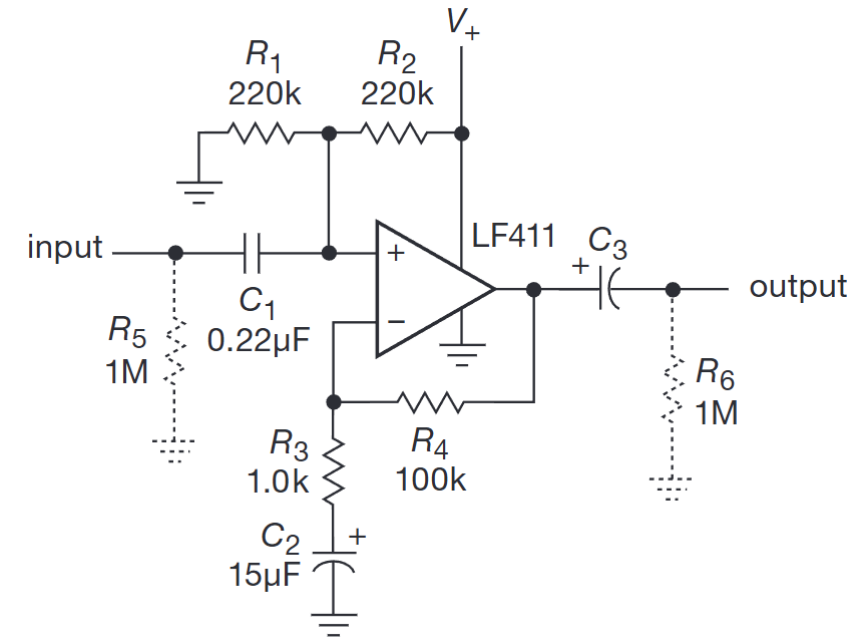


Fig. 3. Biased circuit to $V_{ref} = \frac{1}{2}V_+$

Solution – Additional Components of the Biased Circuit

- C1: AC coupling on the input, blocking DC (same as Fig. 1)
- C2: DC $\rightarrow$ open circuit, AC $\rightarrow$ short circuit.
 - AC ground, stabilizes the gain at high frequencies
- C3: AC coupling on the output
 - Blocks 6 V DC offset into the next stage and lets only the AC audio signals pass through
 - Ensures the output is centered at 0 V externally
- R5, R6 (optional):
 - Ensures C1 (input) and C3 (output) have defined DC paths
 - Prevent floating nodes, random DC offsets, or noise
- R3, R4: closed-loop negative feedback (same as Fig. 1)
- Note: C2 & C3 $\rightarrow$ polarized due to large μF value + fixed DC polarity

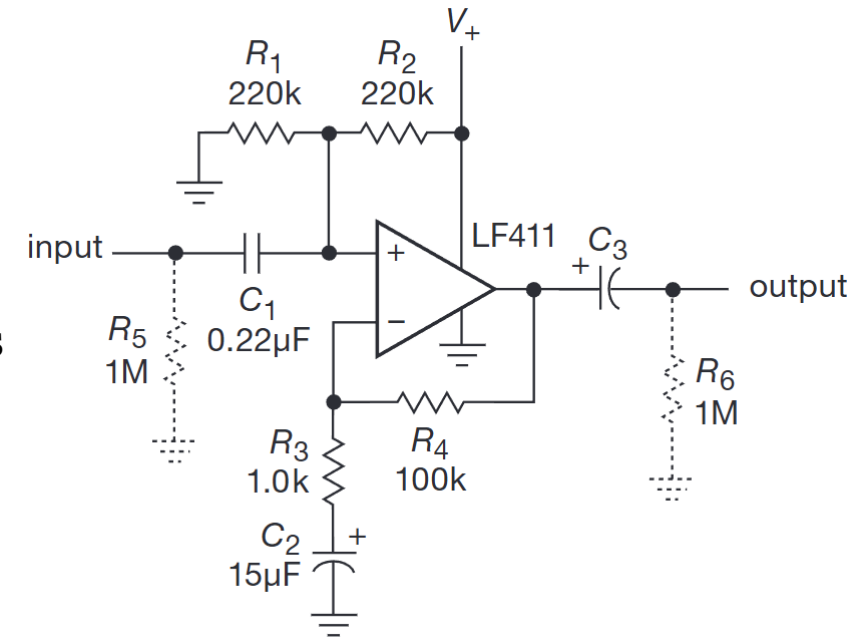


Fig. 3. Biased circuit to $V_{ref} = \frac{1}{2}V_+$

Simulation - Defective

- [Defective Falstad simulation link](#)

Simulation - Solution

- [Fixed Falstad simulation link](#)

Articulate Learning

Recognize the need for learning / continued learning:

- Lack of knowledge on AC coupling
- Audio signals – AC vs DC

Self-Directed Learning:

- Research on audio signal amplifier behaviour
- Textbook, LF411 op-amp datasheets
- Experimented with Falstad simulations circuits, component values, and graphs

Articulate Learning

Technical Learning:

- Real-life implications of audio signaling
- Biasing circuits and reference voltages
- Separating DC bias from AC signal behaviour

I will apply these biasing techniques and what I learned overall through this project in my future work :)

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